A Conditional Weakening of GCH in the Koszmider–Shelah–Swietek Construction

arXiv automatic math run `fa_banach_001`

Candidate conditional packet, June 14, 2026

Classification

- **Status:** conditional reduction.
- **Source:** P. Koszmider, S. Shelah, M. Swietek, *There is no bound on sizes of indecomposable Banach spaces*, arXiv:1603.01753; Adv. Math. 323 (2018), 745–783.
- **Target:** the Introduction asks whether the generalized continuum hypothesis can be removed from Theorem 1.3, which under GCH produces, above every cardinal, an indecomposable $C(K)$-space with few operators.

Source Question

On page 5, immediately after the repeated Theorem 5.3, the source paper asks whether the GCH hypothesis can be removed from Theorem 1.3. In the parsed source, this is line 396 of `source.tex`.

> needed to obtain Theorem 1.3 is:
>
> **Theorem 5.3.** *Assume the generalized continuum hypothesis. Let κ be the successor of a cardinal of uncountable cofinality. There is a compact Hausdorff connected c.c.c. space K of weight κ without a butterfly point such that $C(K)$ has asymmetric distribution of separations in the direction of some $\mathcal{D} \subseteq C_I(K)$.*
>
> We do not know if the hypothesis of the generalized continuum hypothesis can be removed from Theorem 1.3. In [37] Shelah's black boxes were used to avoid any additional set theoretic assumption in the construction of endo-rigid Boolean algebras. The Banach space construction seems more demanding in this context. The first and the third named authors would like to thank Gabriel Salazar for discussions concerning Shelah's black boxes.
>
> The obtained spaces $C(K)$ have other usual properties of $C(K)$s with few operators proved in [21] such as having proper subspaces, in particular hyperplanes

Conditional Dependencies

This packet does *not* prove the full ZFC version of the source question. It isolates two assumptions which are sufficient for the source construction at a fixed cardinal.

Let κ be a regular cardinal which is the successor of a cardinal of uncountable cofinality. The dependencies are:

(H1) $\kappa^\omega = \kappa$ and $\alpha^\omega < \kappa$ for every $\alpha < \kappa$.

(H2) $\diamondsuit(E^\kappa_\omega)$: there is a sequence $(S_\alpha)_{\alpha\in E^\kappa_\omega}$ such that $S_\alpha \subseteq \alpha$ and, for every $X \subseteq \kappa$, the set

$$\{\alpha \in E^\kappa_\omega : X \cap \alpha = S_\alpha\}$$

is stationary in κ.

Under GCH, the source obtains (H1) from Lemma 5.1 and (H2) from Gregory's theorem, quoted as Theorem 5.2. The point of the packet is that the rest of the argument uses only these two consequences.

Proof Intuition

The source proof uses GCH as infrastructure, not as a Banach-space ingredient. Countable cardinal arithmetic keeps the Boolean completion, the function families, and the possible countable codes at size κ, and also makes the coding-closure set club. The diamond sequence then guesses the eventual countable data needed at stationarily many stages of the ladder recursion. Once these two ingredients are assumed directly, every later topological and operator-theoretic step is exactly the source argument: construct the ladder family, obtain asymmetric distribution of separations, deduce few-star operators from the c.c.c. space, upgrade to few operators using absence of butterfly points, and conclude indecomposability from connectedness.

Conditional Theorem

Theorem 1. *Let κ be a regular cardinal which is the successor of a cardinal of uncountable cofinality. Assume (H1) and (H2) above. Then there is a compact Hausdorff connected* c.c.c. *space K of weight κ, without a butterfly point, such that $C(K)$ has asymmetric distribution of separations in the direction of some $\mathcal{D} \subseteq C_I(K)$. Consequently, $C(K)$ has few operators and is indecomposable, and* $\operatorname{dens} C(K) = \kappa$.

Proof. We verify that the proof of Theorem 5.3 of the source paper works with (H1) and (H2) in place of GCH.

First, the source Lemma 3.13, labelled "GCH" in the paper, proves that $\operatorname{dens} C(\nabla\mathcal{F}) = \kappa$ for the relevant $\mathcal{D} \subseteq \mathcal{F} \subseteq C_I(L)$. Inspecting its proof, the only cardinal calculation is that the completed free Boolean algebra $\overline{Fr}(\kappa)$ has cardinal at most κ^ω, because every element is the supremum of a countable subset of the free algebra $Fr(\kappa)$. Thus the proof needs only $\kappa^\omega = \kappa$, which is part of (H1). The lower bound comes from the coordinate family $\mathcal{D}$, and the upper bound comes from the Stone-Weierstrass estimate inside $C(L)$, as in the source proof.

Second, at the beginning of Section 5, the source Lemma 5.1 is used only for the two displayed conclusions

$$\kappa^\omega = \kappa \quad \text{and} \quad \alpha^\omega < \kappa \quad (\alpha < \kappa).$$

These are exactly (H1). They give the size of the coding space

$$(\{-2,-1\} \cup \kappa) \times \{-1,1\}^{\mathbb{N}} \times \left(C_I(L)^{\mathbb{N}} \cup \mathcal{B}(C_I(L))^{\mathbb{N}}\right)$$

as κ, so the bijection Ψ used by the construction exists. They also supply the standard closure argument showing that the set C_Ψ of ordinals closed under the coding map is club in κ.

Third, the only use of source Theorem 5.2 is to obtain a $\diamondsuit(E^\kappa_\omega)$-sequence. This is exactly (H2). With that sequence fixed, the recursive definition of the ladder family $\mathcal{F}$ is verbatim the construction in Theorem 5.3. At a stage $\gamma < \kappa$, the existence of the required $\alpha_\gamma \in E^\kappa_\omega \cap C_\Psi$ follows from (H2), because the target code $X \subseteq \kappa$ is guessed on a stationary set and C_Ψ is club. The rest of the stage uses only the source lemmas on ladder families, countable separating subfamilies, and nonseparation along an infinite $M \subseteq \mathbb{N}$.

The verification of asymmetric distribution of separations is also unchanged. Given pairwise disjoint (f_n), antichains (U_n), signs (ν^ξ_n), and refinements (U^ξ_n), code all of this by a subset of κ. By (H2) choose a guessed $\alpha \in E^\kappa_\omega \cap C_\Psi$ above a bound β containing the supports of the data. The ladder family stage at α supplies

$$g_\alpha = \bigvee_{n \in M_\alpha} \left(f_n d_{\eta^\alpha_n, \nu^{\eta^\alpha_n}_n} \right)$$

and the nonseparation conclusion required in Definition 1.5 of the source paper. Thus $C(\nabla\mathcal{F})$ has asymmetric distribution of separations in the direction of $\{d_\alpha : \alpha < \kappa\}$.

The source lemmas on ladder families then give that $K = \nabla\mathcal{F}$ is connected and has no butterfly point; the source c.c.c. argument for L_κ and its continuous image gives that K is c.c.c.; and the density computation above gives weight and density κ. Applying source Theorem 2.1 and Theorem 2.4, asymmetric distribution plus c.c.c. gives few-star operators, no butterfly points upgrades this to few operators, and connectedness makes $C(K)$ indecomposable. □

Corollary 1. *If for every cardinal μ there is a regular successor cardinal $\kappa > \mu$, with predecessor of uncountable cofinality, satisfying (H1) and (H2), then the conclusion of source Theorem 1.3 holds without assuming full* GCH*: for every μ there is an indecomposable Banach space of density bigger than μ, of the form $C(K)$ with few operators and connected compact K.*

Proof. Apply the theorem to such a $\kappa > \mu$. The resulting space has density κ, hence density bigger than μ, and has the same few-operators and indecomposability conclusion as in the source theorem. □

Verification and Novelty Notes

The proof is a line-by-line dependency audit of the source construction, not an independent reconstruction of the ladder family. The critical source locations are:

- Lemma 3.13, source lines 1116–1130, where $\kappa^\omega = \kappa$ is used for the density calculation.
- Lemma 5.1, source lines 2025–2036, whose conclusions are absorbed into (H1).
- Theorem 5.2 and the setup before Theorem 5.3, source lines 2043–2086, where GCH is used to obtain $\diamondsuit(E^\kappa_\omega)$ and the coding map Ψ.
- Theorem 5.3, source lines 2088–2242, where the recursion and the final asymmetric-distribution verification use the diamond sequence and the coding club.

Bounded duplicate and literature checks were performed before promotion. The run registry, solution, attempt, and proof-gap indexes had no hit for arXiv:1603.01753, the exact title phrase, or the GCH-removal question. Local corpus searches for the title/authors together with *GCH*, diamond, black-box, few-operator, and arbitrary-density terms did not reveal a later exact answer. A focused web search for the exact title and phrases such as "GCH removed" and "generalized continuum hypothesis removed" found the source paper but no separate paper explicitly removing GCH from the arbitrary density few-operators construction. The result is therefore recorded only as a candidate conditional reduction, not as a full solution of the ZFC removal problem.

Limitations

The packet leaves the main source question open in its full form. In particular, it does not prove $\Diamond(E^\kappa_\omega)$ in ZFC, does not prove that the countable-coding arithmetic in (H1) holds above every cardinal, and does not replace Shelah black boxes by a new Banach-space construction. It only shows that full GCH is not used beyond the two isolated dependencies.

Human Review Recommendation

Reviewers should first check that Lemma 3.13 has no hidden use of GCH beyond $\kappa^\omega = \kappa$, and that the clubness of C_Ψ indeed follows from $\alpha^\omega < \kappa$ for all $\alpha < \kappa$. The next useful mathematical step would be either to establish (H1)+(H2) in a weaker set-theoretic context relevant to many cardinals, or to replace $\Diamond(E^\kappa_\omega)$ in the ladder recursion by a genuine black-box construction.

References

[1] P. Koszmider, S. Shelah, and M. Swietek, *There is no bound on sizes of indecomposable Banach spaces*, arXiv:1603.01753; Adv. Math. 323 (2018), 745–783.

[2] T. Jech, *Set Theory*, Springer Monographs in Mathematics, Springer, 2003.